Scales of Nature

<u>Synopsis</u>: We explore objects in the universe from very small to extremely large. Because the range of numbers spanned is unimaginably large, we describe the sizes in terms of orders of magnitude, i.e., the number of zeros in the number.

<u>Performance Expectations</u>: Describe quantities in order-of-magnitude scales.

<u>Focus on Science Practices</u>: Using mathematics and computational thinking, Scales, proportion, and quantity.

<u>Instructions and Questions</u>

1) Go to the website https://scaleofuniverse.com/en (or search for the scale of the universe on Google)
2) For each object below, give an order-of-magnitude size in meters.
   (Write $10^{\square}$, you may give a range, $10^{\square}$ - $10^{\square}$)

Object	**Order-of-magnitude size in meters**
Quark	
Nucleus	
Atom	
Small molecule (a few atoms)	
Large molecules (e.g., proteins, 10s of thousands of atoms)	
Cell	
Human height	
Radius of the Earth	
Radius of a large planet (e.g., large ones in the solar system)	
Size of Solar System (Kuiper Belt)	
Distance to a nearby star (Proxima Centauri)	
Size of the Milky Way Galaxy	
Size of clusters of galaxies	
Size of superclusters of galaxies	
Size of the observable universe	

Q1) By how many orders of magnitude (powers of 10) is the largest larger compared to the smallest? (Hint: Take the ratio of the large to small = difference in the number of zeros.)

Q2) How many would span a typical human if you laid the following from end to end?

i) Atom:

ii) Cell:

3) For each object below, give an order-of-magnitude size in kg.
(Write $10^{\square}$, you may give a range, $10^{\square}$ - $10^{\square}$)

Object	**Order-of-magnitude size in kg**
Electron	
Proton	
Cell	
Grain of sand	
Drop of water	
Cell phone	
Human	
Compact car	
Everest mountain	
All liquid water on Earth	
Moon	
Earth	
Sun	
Black hole at the center of the Milky Way	
Milk Way	
Universe	

Q3) What is the ratio of the largest to the smallest mass?

Q4) If the universe's mass is the total of the mass of all galaxies (the Milky Way is a galaxy, and we belong here), roughly how many galaxies make up the universe? Show calculation.

Q5) If the mass of a galaxy is roughly the total mass of all stars (the Sun is a star), roughly how many stars make up a galaxy? Show calculation.

Q6) Based on your calculations in Q4 and Q5, roughly how many stars are in the universe?

Q7) Roughly how many drops of water make up all the liquid water on the Earth?